EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Spiral development is simply waterfall repeated without evaluating project risks.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

## Lesson 119 · Spiral Development and Method Selection

Name: _____ Class: _____ Date: _____

**1. What is one distinguishing XP practice?**

**2. An OR gate has inputs 0 and 1. What is the output?**

**3. What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?**

**4. What is the time complexity of sequential search in the worst case?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

For example TDD, pair programming, CI or frequent refactoring.

**2. Model answer**

1.

**3. Model answer**

X is removed; Y remains at the front.

**4. Model answer**

O(n).

**By the end, you can…**

- Set objectives, options and constraints for a spiral iteration.
- Evaluate and reduce technical or organisational uncertainties before commitment.
- Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

**Vocabulary — know these terms**

spiral model — Risk-driven iterative development lifecycle.

risk analysis — Estimating likelihood and impact of potential problems.

prototype — Early model built to explore or test an idea.

iteration — Repeated cycle of development and evaluation.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Compare waterfall, agile/RAD, XP and spiral for a high-risk changing specification. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

Describe three central activities within one spiral development cycle. [6]

**Question 2**

Give two reasons a fixed-requirement small project might not benefit from the spiral model. [6]





NEW LEARNING

# Read, annotate and explain

**Plan the cycle**

Set objectives, options and constraints for a spiral iteration.

**Analyse risks**

Evaluate and reduce technical or organisational uncertainties before commitment.

**Prototype and review**

Develop/verify increment, consult stakeholders, plan next risk-driven cycle.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A bank is building a novel payment verification system with security uncertainty. Justify the spiral model.**

### Model answer

Each cycle sets goals, identifies risks such as fraud or performance, prototypes/assesses the riskiest elements, evaluates with stakeholders then plans the next cycle. Risk reduction drives the project.

Check: link each claim to the scenario and use precise OCR terminology.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve